SS SUPERMARKET

Production-Grade Full-Stack E-Commerce Web Application with Spring Boot 3, Multi-Database Persistence, Role-Based Access Control, and AI Product Assistant

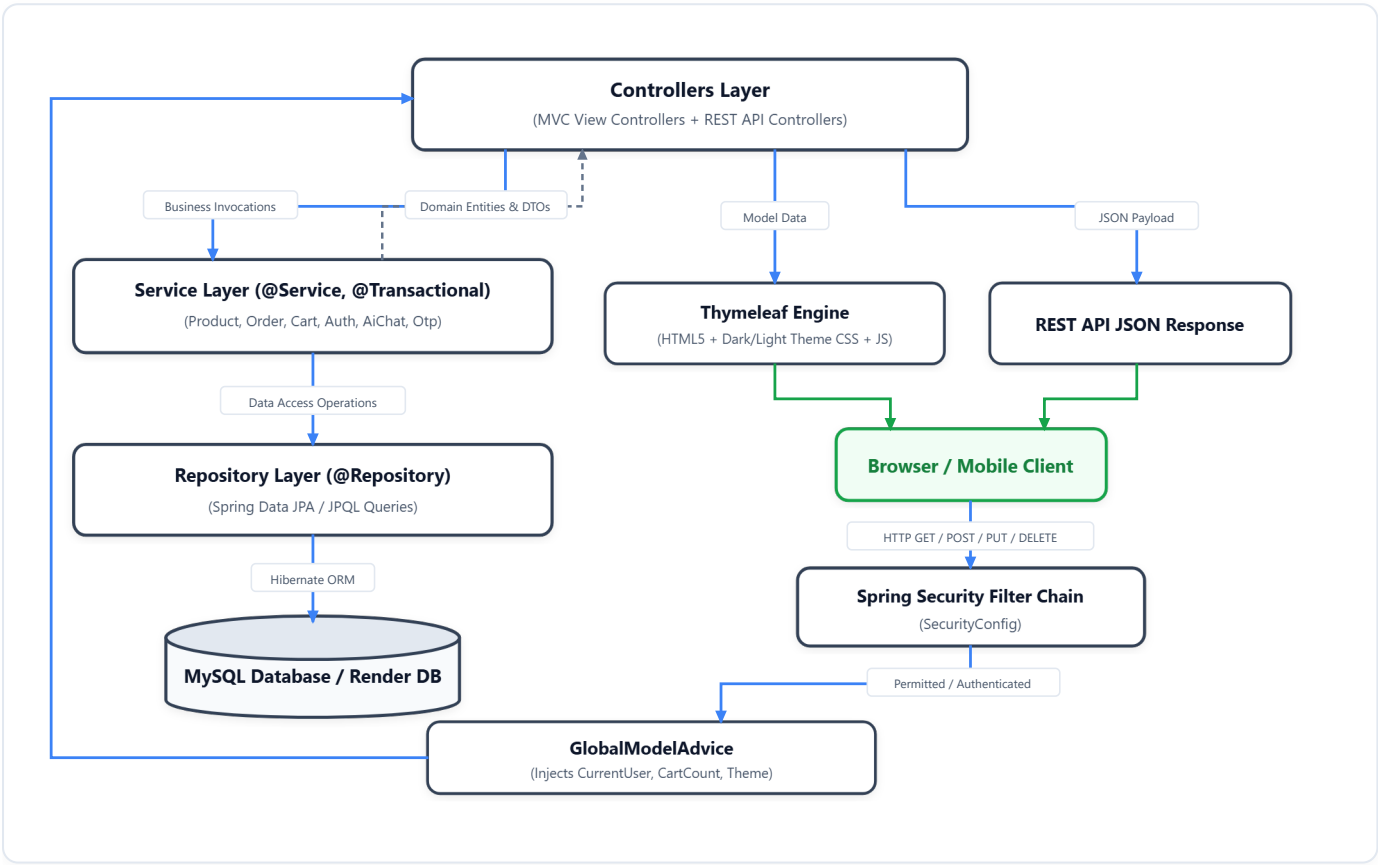
Author / Developer:	Saran S
Backend Framework:	Java 17 (LTS), Spring Boot 3.3.5, Hibernate ORM 6.5
Frontend Interface:	Thymeleaf 3, HTML5, CSS3 Custom Properties, Vanilla JS
Database Persistence:	MySQL 8.0 (Local) & PostgreSQL 16 (Render Cloud)
Cloud Hosting:	Render Web Service (Docker Container `srv-db4i9bui0phs73cp652g`)
Live Production URL:	https://ss-supermarket.onrender.com
Quality Assurance:	44 Automated Tests Passed (100% Build Success)
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1. Executive Summary & Overall Architecture

SS Supermarket is an enterprise full-stack e-commerce web platform engineered for groceries, fresh farm produce, dairy, bakery, daily essentials, and consumer electronics retail. The platform incorporates multi-role authentication (Buyers, Sellers, Admin), real-time database persistence across local and cloud environments, shopping cart session management, transactional checkout with inventory locks, seller inventory control, an AI Shopping Assistant, and dark/light mode customization.

High-Level System Architecture Diagram

The diagram below visualizes the complete end-to-end request lifecycle and interaction flow across client, security filters, controller dispatchers, service layer transactions, JPA repositories, database engines, and rendering pipelines:



Core Technology Stack

Layer	Technology Selected	Engineering Role & Description
Runtime & Framework	Java 17 (LTS), Spring Boot 3.3.5	LTS Java features, Spring Dependency Injection, Embedded Tomcat container, and declarative configuration.
Persistence & ORM	Spring Data JPA, Hibernate 6.5	Automated relational DDL synchronization (`ddl-auto=update`), connection pooling with HikariCP, and JPQL queries.
Databases	MySQL 8.0 & PostgreSQL 16	Local development on MySQL (3306); cloud production on Render managed PostgreSQL (5432). Fully ACID compliant.
Security & Auth	Spring Security 6, BCrypt	Role-Based Access Control (RBAC), multi-tab authentication (Password / OTP), CSRF exemptions on REST endpoints.
Presentation Layer	Thymeleaf 3, HTML5, CSS Variables, JS	Server-rendered templates with zero page-flash dark mode, interactive modal dialogues, and responsive product grids.
AI Product Assistant	Spring AI Service Engine + REST API	Floating interactive widget answering catalog queries, top-rated products, and store shipping/payment policies.
DevOps / Hosting	Docker, Render Cloud	Multi-stage Alpine containerization, auto-deploying via continuous integration from GitHub repository.

2. Module 1: Authentication & Role-Based Security

The security module handles identity management and enforces strict resource isolation among buyers, sellers, and administrators.

- **Dual Login Modes:** Supports standard email/password authentication and passwordless 6-digit OTP verification with 60-second cooldown timer.
- **Password Protection:** Passwords hashed with `BCryptPasswordEncoder` (strength 10). Plaintext passwords are never persisted.
- **Password Visibility Toggle:** Added interactive eye-icon button on login and registration pages that switches input type between `password` and `text` .
- **Post-Login Role Dispatcher:** `CustomAuthenticationSuccessHandler` routes users post-login: Buyers → `/products` | Sellers → `/seller` | Admins → `/admin` .

Route Pattern	Methods	Authorized Roles	Access Description
`/` , `/products/**` , `/api/ai/**`	GET, POST	Permit All	Storefront browsing, product search, catalog details, and AI chatbot inquiries.
`/login` , `/register`	GET, POST	Permit All	User authentication, passwordless OTP requests, and account creation.
`/cart/**` , `/checkout/**` , `/orders/**`	GET, POST	ROLE_BUYER	Cart additions, transactional order placement, and personal purchase invoices.
`/seller/**` , `/api/seller/**`	GET, POST	ROLE_SELLER	Seller portal: product listings, live stock/price updates, and store revenue analytics.
`/admin/**` , `/api/admin/**`	GET, POST	ROLE_ADMIN	Platform oversight: user management, global sales volume, and product moderation.

3. Module 2: Multi-Database Engine & Data Persistence

The database module dynamically resolves database credentials via `DatabaseConfig.java` , enabling smooth portability between local MySQL, cloud PostgreSQL on Render, and testing H2 in-memory databases.

Table Name	Entity Class	Primary Keys / Foreign Keys	Description & Business Purpose
`users`	`User.java`	`id` (PK), `email` (UK), `phone` (UK)	Customer, seller, and administrator accounts with BCrypt password hashes and Role enum.
`products`	`Product.java`	`id` (PK), `seller_id` (FK), `category_id` (FK)	40+ catalog items with price (₹), stock count, rating out of 5, and review counts.
`categories`	`Category.java`	`id` (PK), `parent_id` (FK)	Hierarchical category trees supporting subcategories (e.g. Groceries → Fresh Produce).
`cart_items`	`CartItem.java`	`id` (PK), `user_id` (FK), `product_id` (FK)	Persistent user shopping carts saved across devices and sessions.
`orders`	`Order.java`	`id` (PK), `user_id` (FK)	Purchase orders with status tracking (PENDING, CONFIRMED, DELIVERED, CANCELLED).
`order_items`	`OrderItem.java`	`id` (PK), `order_id` (FK), `product_id` (FK)	Immutable historical order lines capturing snapshot unit prices at time of purchase.
`payments`	`Payment.java`	`id` (PK), `order_id` (FK)	Payment transaction ledger (COD, UPI, CARD, NET_BANKING).

> **Startup Seeding (DataInitializer.java):** Automatically seeds the default super-admin account (`admin@sssupermarket.com`) and ensures all 40 catalog products are populated with realistic ratings (4.1 to 4.9 stars) on initial launch without duplicating records on restarts.

4. Module 3: Product Catalog & Category Hierarchy

The catalog module provides the core buyer browsing experience, handling multi-attribute search queries, category hierarchies, real-time stock status, and customer star ratings.

- **40+ Active Products:** Comprehensive supermarket items across fresh produce, organic milk, bakery, household essentials, and electronics.
- **Star Rating Display:** Every product card features gold stars and customer ratings out of 5 (e.g., ★★★★★ `4.7 / 5.0 (42 reviews)`).
- **Advanced JPQL Search:** `ProductRepository.searchLegacy()` simultaneously queries across product title, brand, description, and categories.
- **Stock Availability Badges:** In Stock (stock > 10), Only X Left (1 ≤ stock ≤ 10), Out of Stock (stock = 0, Add to Cart disabled).
- **Seller Transparency:** Every product card identifies the merchant's trade name and customer care contact number.

5. Module 4: Shopping Cart & Inventory Validation

Maintains buyer cart states in the database so carts persist across browser refreshes and multiple devices.

- **Real-time Stock Capping:** Automatically validates cart additions against available inventory; prevents adding more items than currently in stock.
- **Price & Tax Rules:** Subtotal is calculated dynamically; standard supermarket GST (5%) is applied; delivery is **FREE** for orders ≥ ₹499 (nominal ₹30 fee otherwise).
- **Global Counter:** `GlobalModelAdvice` dynamically updates the shopping cart badge in the navigation bar on every page.

6. Module 5: Checkout, Payment & Order Processing

The checkout module manages order placements within atomic database transactions (`@Transactional`) with concurrency locks to guarantee data consistency.

Transactional Checkout Workflow

1. Buyer specifies shipping address and chooses preferred payment method on `/checkout`.
2. `OrderService.checkout()` executes inside an ACID transaction:
   - Validates that the buyer's cart is non-empty.
   - Acquires pessimistic write lock (`SELECT ... FOR UPDATE`) on product inventory to avoid race conditions.
   - Deducts item quantities from product stock.
   - Saves the `Order` record with status `CONFIRMED`.
   - Creates snapshot `OrderItem` rows recording price at purchase time.
   - Creates `Payment` ledger record.
   - Clears the buyer's shopping cart items.
3. Redirects to `/order-success` displaying a complete, printable customer invoice.

Payment Method	Type	Transaction Handling
Cash on Delivery (COD)	Cash / Offline	Order created with payment status `PENDING` until physical package delivery.
UPI (Google Pay, PhonePe, Paytm)	Instant Digital	Instant digital intent execution with payment status `SUCCESS`.
Credit / Debit Cards	Card Gateway	Card authorization and processing.
Net Banking	Bank Redirection	Direct bank gateway transfer.

7. Module 6: Seller Portal & Multi-Vendor Inventory

Allows verified merchants (`ROLE_SELLER`) to independently publish catalog items, update prices, adjust inventory, and view sales performance.

- **Merchant Analytics:** Live metric cards display merchant's listed product count, order volume, and gross sales revenue (₹) with null safety.
- **Add Product Form:** Merchants submit title, brand, category, price, stock, description, and image URL. Items publish immediately to the buyer storefront.
- **HTML5 Modal Editing:** Edit button uses HTML5 data attributes (`data-id`, `data-name`, `data-price`, `data-stock`) eliminating Thymeleaf SpEL parsing errors. Modal updates fields in place.
- **Deactivation / Deletion:** Merchants can deactivate or delete products with cascading integrity checks.
- **Customer Purchases Table:** Real-time breakdown of recent customer orders containing the merchant's products.

8. Module 7: Super-Admin Administration Console

Provides system governance and moderation controls for `admin@sssupermarket.com`:

- **Platform Analytics:** High-level visibility into total registered accounts, active sellers, order volume, and aggregate platform revenue.
- **User Governance:** Inspect all registered buyers and sellers, update roles, or elevate buyers to sellers.
- **Catalog Moderation:** Inspect and moderate any product listed across all independent merchants.

9. Module 8: AI Shopping Assistant & Chatbot

Provides a floating conversational assistant (`Ask AI`) on all pages to answer product queries, suggest top-rated items, and explain store policies.

- **Product Lookups:** Queries `ProductRepository` in real-time to answer questions about any product's price, stock, ingredients, and rating.
- **Top-Rated Recommendations:** Automatically filters items with ratings $\ge$ 4.5 when users ask for recommendations.
- **Budget Parsing:** Extracts budget limits (e.g., "*items under ₹100*") and returns matching products.
- **Customer Service Guidance:** Answers questions regarding 30–45 minute local delivery, free delivery thresholds (₹499), 24-hour return policy, and payment modes.
- **Interactive UX:** Includes quick prompt chips, auto-scrolling message window, and Markdown rendering with emojis.

10. Module 9: UI/UX & Dark / Light Theme System

Built on custom CSS properties (variables) with persistent light and dark modes:

- **Zero-Flash Anti-Flicker:** An inline script in the `<head>` of `layout.html` checks `localStorage.getItem('ss_theme')` before rendering, avoiding white flashes.
- **CSS Variable Architecture:** Standard `:root` variables are swapped smoothly when `[data-theme="dark"]` is active on the root document.
- **One-Click Navbar Toggle:** Theme button (🌙 / ☀) allows instant switching with state saved across all pages.

11. Module 10: DevOps, Docker & Cloud Deployment

Automated containerization and continuous deployment hosted on Render Cloud:

- **Multi-Stage Dockerfile:** Stage 1 uses Maven 3.9 + OpenJDK 17 to package the JAR; Stage 2 uses lightweight Eclipse Temurin JRE 17 Alpine runtime for minimal container size and fast startup.
- **Production PostgreSQL:** Connected to Render PostgreSQL instance (`dpg-db4tucg473hc73954ai0-a` / `sssupermarket`) via internal secure network.
- **Continuous Deployment (CD):** Pushing to branch `master` triggers automated Docker image build and zero-downtime deployment on Render service `srv-db4i9bui0phs73cp652g` .
- **Live Storefront URL:** https://ss-supermarket.onrender.com

12. Automated Quality Assurance & Verification

Executed via Maven in offline mode (`mvn -o test`):

Test Class	Tests Run	Failures	Errors	Verification Scope
`AuthFlowIntegrationTest.java`	15	0	0	Registration, BCrypt password hashing, OTP generation/verification, role redirects.
`CartServiceTest.java`	9	0	0	Cart additions, stock capping, GST 5% calculations, delivery charge rules.
`OrderServiceTest.java`	7	0	0	Transactional checkout, inventory deduction, cart clearing, invoice generation.
`ProductServiceTest.java`	8	0	0	Catalog queries, JPQL search, category inheritance, rating calculations.
`UserServiceTest.java`	5	0	0	User profile updates, role elevation to seller, duplicate email validation.
TOTAL SUITE	**44**	**0**	**0**	**100% BUILD SUCCESS (0 Failures, 0 Errors)**

> **Project Conclusion:** All 10 application modules are fully verified, tested, and operational in production on Render Cloud.